

Sarthik Vaghasiya

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LinkedIn | Portfolio

SUMMARY

Full-Stack Developer with over 2 years of experience building cross-platform mobile applications (Flutter), Angular web platforms, and scalable Node.js backend APIs. Specialized in real-time systems using WebRTC, Socket.IO, and MQTT, with a strong track record of delivering production-grade applications on Play Store and App Store. Proven ability to architect event-driven, low-latency systems across both mobile and web, with measurable improvements in performance and scalability.

EXPERIENCE

Full Stack Engineer

Jenex Technovation Pvt Ltd | April 2024 – Present

- Implemented event-driven real-time communication using MQTT and Socket.IO, reducing latency by 40% and improving device responsiveness across distributed clients
- Built a low-latency, scalable video streaming system supporting up to 8 concurrent live feeds using WebRTC and AWS Kinesis, ensuring stable real-time video delivery
- Implemented real-time GPS tracking using Google Maps, optimizing data processing and update frequency to improve location consistency and reduce unnecessary re-renders
- Developed Angular web dashboards for real-time data visualization, device management, and role-based access control across multiple product lines
- Published multiple mobile applications on Play Store and App Store, ensuring high performance, stability, and compliance with platform guidelines
- Contributed to real-time, event-driven system architecture, reducing dependency on polling mechanisms and improving backend efficiency and scalability
- Developed cross-platform mobile applications with BLE and Wi-Fi integration for real-time IoT device control, monitoring, and seamless data synchronization

TECHNICAL SKILLS

- Languages: Dart, JavaScript, TypeScript
- Mobile Development: Flutter (Riverpod, GetX)
- Frontend (Web): Angular
- Backend Development: Node.js, Express.js, REST APIs
- Databases: MongoDB, MySQL
- Real-time & Messaging: WebSockets, MQTT, Socket.io, WebRTC
- Tools & Platforms : Firebase, CI/CD, Git

KEY PROJECTS

Real-time Tracking & Monitoring Platform - Flutter (Riverpod), Socket.io, MQTT, Google Maps SDK, Node.js, Angular

- Built scalable real-time tracking system with live map updates, geofencing, alerting, and telemetry processing
- Migrated from timer-based polling to event-driven architecture using Socket.io and MQTT
- Reduced unnecessary API calls by 80%, significantly improving system performance
- Optimized map rendering by implementing data caching and controlled state updates, reducing UI lag
- Designed dynamic UI architecture supporting multiple product versions within a single application
- Developed a corresponding Angular web application providing the same real-time tracking, geofencing, and telemetry features for web-based users

Real-time Communication & Device Control Platform - Flutter (GetX), BLE

- Built a Bluetooth Low Energy (BLE)-based communication system for offline marine environments with limited or no network connectivity
- Developed a cross-platform Flutter application for real-time device control and monitoring via BLE communication
- Designed a reliable offline communication workflow, ensuring continuous operation without internet dependency
- Built alarm management features with automated and manual triggers for safety-critical scenarios

Live Video Streaming System - WebRTC, AWS Kinesis, Flutter, Angular

- Built a low-latency video streaming pipeline integrated into mobile applications for real-time video delivery
- Supported up to 8 concurrent camera feeds with stable performance across devices
- Implemented real-time video communication using WebRTC, ensuring minimal latency and smooth playback
- Integrated AWS Kinesis for scalable video ingestion and stream distribution
- Developed multi-feed stream handling and switching logic for seamless user experience
- Built a companion Angular web application enabling browser-based access to live camera feeds with the same multi-feed switching and role-based stream controls

E-Commerce Application - Node.js, Angular, Express.js, MongoDB, JWT

- Built a full-stack e-commerce platform with responsive design, optimized purchase flow, and product management including category filtering, inventory tracking, and order processing
- Implemented JWT-based authentication and RBAC for Admin, Seller, and Buyer roles
- Optimized filtering, cart, and checkout experiences to reduce bounce rate and improve conversions

EDUCATION

Bachelor of Engineering (B.E.) - A.D.Patel Institute of Technology, Anand (2020 – 2024)